

BUILD YOUR PERSONAL CYBERSECURITY LAB

**Sumerix Global Virtual Internship – Cybersecurity & Ethical Hacking
Project Task-2: Build Your Personal Cybersecurity Lab**

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1. INTRODUCTION

Cybersecurity professionals require a safe and isolated environment to practice ethical hacking, vulnerability assessment, penetration testing, traffic analysis, and incident response activities. A cybersecurity lab helps students and professionals gain practical experience without affecting real-world systems.

The purpose of creating a personal cybersecurity lab is to safely simulate attacks and defenses in a controlled environment. In this project, a personal cybersecurity lab was developed using VirtualBox, Kali Linux, and OWASP Juice Shop, along with networking concepts such as NAT and Host-Only Adapter.

This lab environment allows secure testing of cybersecurity tools including Nmap, Burp Suite, and Wireshark for scanning, traffic interception, and packet analysis.

2. OBJECTIVE

The objective of this project is to build a safe and isolated cybersecurity practice environment on a personal computer. This lab serves as a foundation for future cybersecurity activities such as:

- Vulnerability Assessment
- Web Application Security Testing
- Penetration Testing Practice
- Network Traffic Analysis
- Incident Response Practice

The lab environment was designed to include:

Components Created	Purpose
Kali Linux	Attacker Machine
OWASP Juice Shop	Vulnerable Target Application
NAT Network	Internet Access

Components Created	Purpose
Host-Only Adapter	Internal Lab Communication

3. LAB ARCHITECTURE

The cybersecurity lab architecture consists of a host machine running VirtualBox virtualization software. Inside VirtualBox, Kali Linux was deployed as the attacker machine, while OWASP Juice Shop was configured as the intentionally vulnerable web application target.

The network architecture was configured using:

NAT Adapter

Used for:

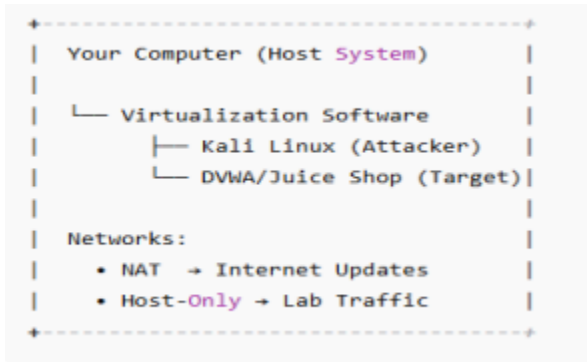
- Internet connectivity
- Software installation
- System updates

Host-Only Adapter

Used for:

- Internal communication within the lab
- Secure isolated testing
- Nmap scanning and traffic analysis

Lab Architecture Diagram



Example Architecture:

Host Computer

- VirtualBox
 - Kali Linux (Attacker)
 - OWASP Juice Shop (Target)
- Networks:
 - NAT → Internet Access
 - Host-Only → Lab Communication

4. PREREQUISITES

The following system requirements were necessary for creating the cybersecurity lab environment:

Requirement	Configuration Used
RAM	24 GB Available
Recommended VM RAM	4 GB Allocated to Kali Linux
CPU	Virtualization Enabled
Disk Space	Sufficient Storage Available
Operating System	Windows

Before starting the setup, virtualization support was confirmed and enabled on the system.

Screenshot Evidence

Virtualization Support Enabled in System



5. TOOLS USED

The following cybersecurity tools and software were used during the lab setup process:

Tool	Purpose
VirtualBox	Virtualization Platform
Kali Linux	Attacker Machine
Docker	Container Deployment
OWASP Juice Shop	Vulnerable Web Application
Nmap	Network Scanning
Burp Suite	Web Traffic Interception
Wireshark	Packet Capture & Analysis
Firefox	Browser Testing

6. STEP-BY-STEP IMPLEMENTATION

Step 1: Installation of VirtualBox

VirtualBox was installed successfully to create and manage virtual machines in an isolated environment.

Procedure

1. Downloaded VirtualBox from official website.
2. Installed VirtualBox using default settings.
3. Verified successful installation.

Step 2: Deployment of Kali Linux Virtual Machine

A prebuilt Kali Linux VirtualBox image was downloaded and imported into VirtualBox.

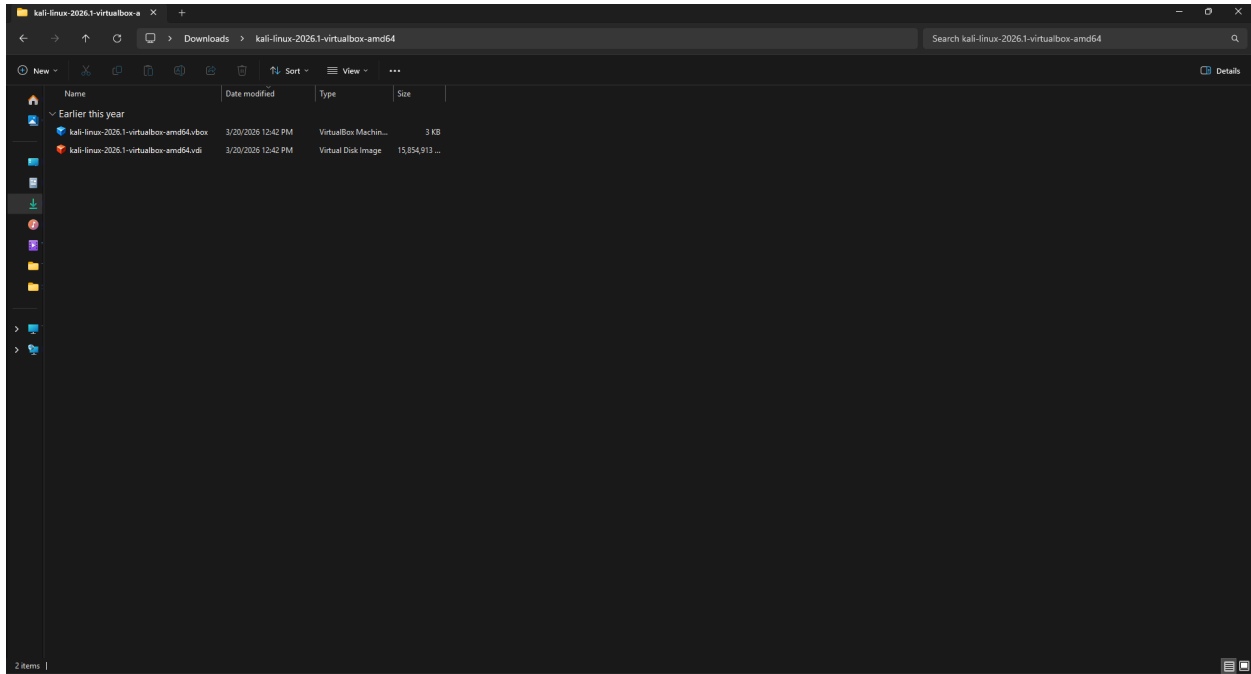
Configuration Applied

Setting	Value
RAM	4096 MB
CPU	2 Processors
Adapter 1	NAT
Adapter 2	Host-Only Adapter

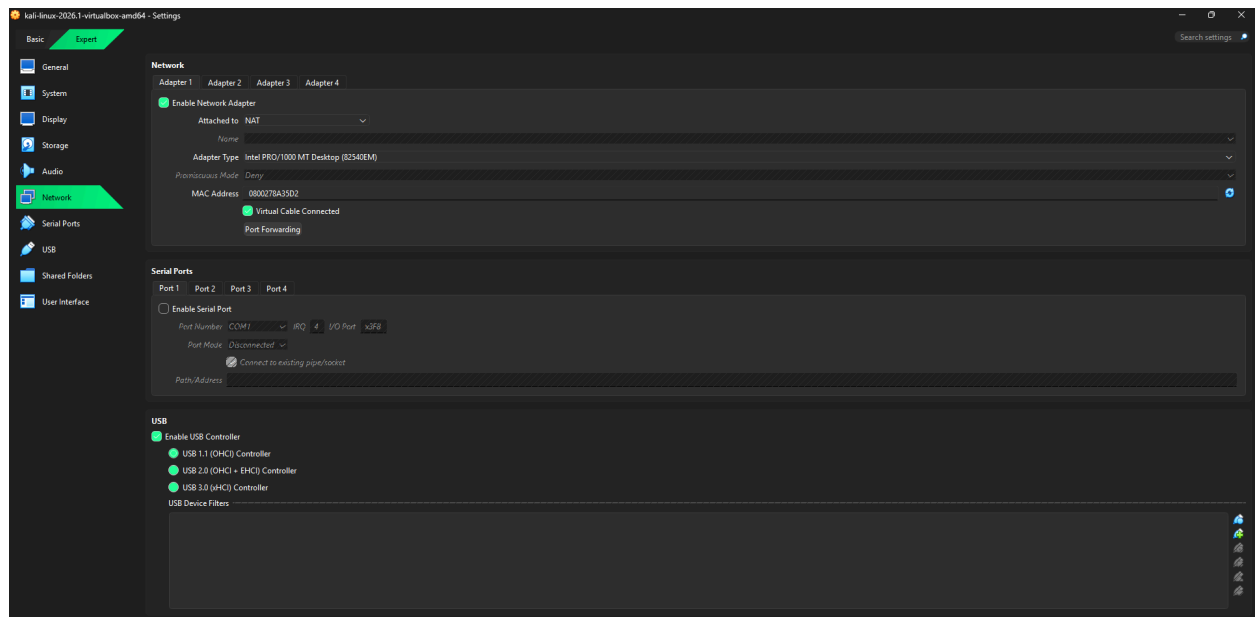
This setup allowed both internet access and isolated internal network communication.

Screenshot Evidence

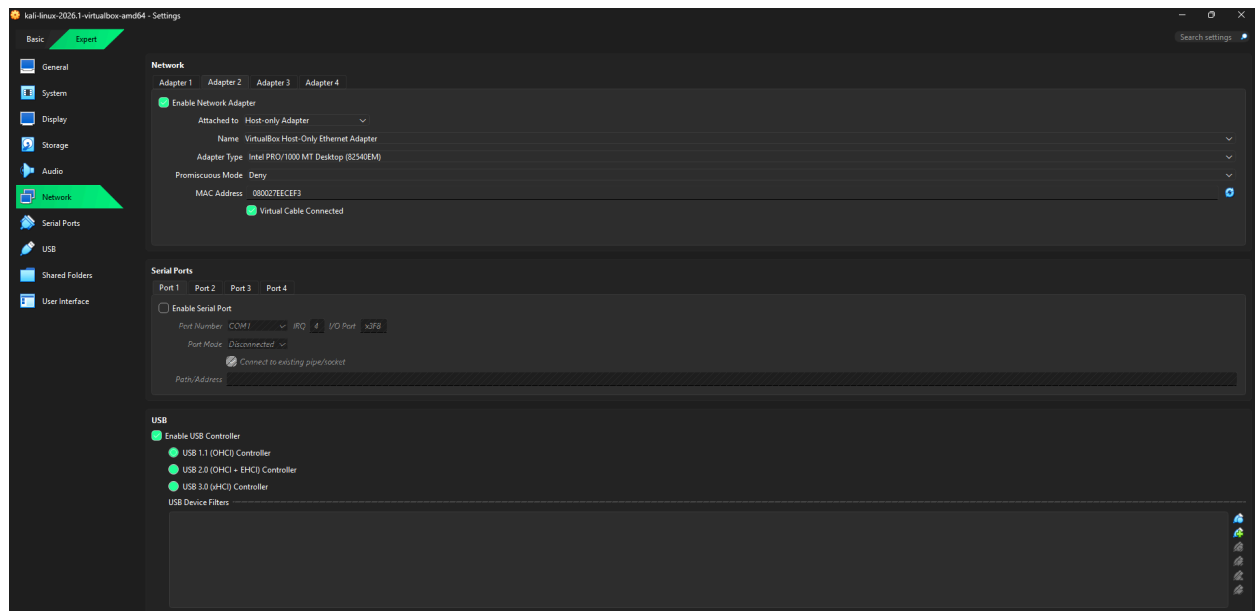
Downloaded Kali Linux Virtual Machine Files



Adapter 1 Configured as NAT



Adapter 2 Configured as Host-Only Network



Step 3: System Update and Lab Folder Creation

After successfully logging into Kali Linux, the system was updated to ensure all packages and repositories were up to date. Updating the system is important for maintaining security, compatibility, and tool functionality.

Commands Used

```
Shell
sudo apt update && sudo apt upgrade -y
```

To maintain proper organization of files and reports during the lab setup process, dedicated directories were created for scans, screenshots, notes, and packet captures.

```
Shell
mkdir -p ~/lab/{scans,notes,pcaps,screenshots}
```

The created folders were verified successfully.

```
Shell
ls ~/lab
```

Purpose of Created Folders

Folder	Purpose
scans	Store Nmap scan outputs
notes	Store observations and findings
pcaps	Store Wireshark packet capture files
screenshots	Store evidence screenshots

Screenshot Evidence

Kali Linux System Update Completed

```
kali@kali: ~
Session Actions Edit View Help
Running update-sys. This may take some time ... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
This may take some time ... done.
Processing triggers for crackmap-runtime (2.9.6-5.2+b2)...
skipping line: 1
Processing triggers for shared-mime-info (2.4-5+b3)...
Processing triggers for libgtk-4-1:amd64 (4.20.3+ds-3)...
Processing triggers for sgml-base (1.31+nmul)...
Processing triggers for delianutils (5.23+2)...
Processing triggers for base-files (1:2026.1.2)...
Setting up polkitd (127+really127-3kali1)...
Upgrading to this polkitd version requires a reboot, please reboot the system when convenient.
polkit-agent-helper.socket was disabled, enabling it.
Created symlink /etc/systemd/system/sockets.target.wants/polkit-agent-helper.socket' + '/usr/lib/systemd/system/polkit-agent-helper.socket'.
Setting up libopenconnect5:amd64 (9.12-3.3+b1)...
Setting up pcsd (2.5.0-1)...
pcsd.service is a disabled or a static unit not running, not starting it.
Setting up pkexec (127+really127-3kali1)...
Setting up openconnect (9.12-3.3+b1)...
Setting up modemmanager (1:24.2-2+b1)...
Setting up kali-menu (2026.2.6)...
Setting up ettercap-graphical (1:0.8.4.1-2)...
Setting up kali-desktop-core (2026.2.11)...
Setting up kali-desktop-xfce (2026.2.11)...
Processing triggers for ca-certificates-java (20260311)...
done.
Setting up default-jre-headless (2:1.25-77)...
Setting up openjdk-21-jre:amd64 (21.0.11-10-1)...
Setting up openjdk-25-jre:amd64 (25.0.3+9-2)...
Setting up default-jre (2:1.25-77)...
Setting up patator (1.1.0-1)...
Setting up kali-linux-headless (2026.2.11)...
Setting up burpsuite (2026.3.2-0kali1)...
Setting up kali-tools-top10 (2026.2.11)...
Setting up kali-linux-default (2026.2.11)...
Processing triggers for dictionaries-common (1.31.4)...
Processing triggers for ca-certificates (20260223)...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
Processing triggers for initramfs-tools (0.151)...
update-initramfs: Generating /boot/initrd.img-6.19.14-kali-amd64
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.44.6+dfsg-2)...
Processing triggers for php8.4-cgi (8.4.22-1)...
Processing triggers for libapache2-mod-php8.4 (8.4.22-1)...
Processing triggers for libc-bin (2.42-10)...
Processing triggers for ca-certificates-java (20260311)...
done.
kali@kali: ~$
```

Lab Folder Structure Created Successfully

```
kali@kali: ~
Firefox ESR
Browse the World Wide Web
Processing triggers for sgml-base (1.31+nmul)...
Processing triggers for delianutils (5.23+2)...
Processing triggers for base-files (1:2026.1.2)...
Setting up polkitd (127+really127-3kali1)...
Upgrading to this polkitd version requires a reboot, please reboot the system when convenient.
polkit-agent-helper.socket was disabled, enabling it.
Created symlink /etc/systemd/system/sockets.target.wants/polkit-agent-helper.socket' + '/usr/lib/systemd/system/polkit-agent-helper.socket'.
Setting up libopenconnect5:amd64 (9.12-3.3+b1)...
Setting up pcsd (2.5.0-1)...
pcsd.service is a disabled or a static unit not running, not starting it.
Setting up pkexec (127+really127-3kali1)...
Setting up openconnect (9.12-3.3+b1)...
Setting up modemmanager (1:24.2-2+b1)...
Setting up kali-menu (2026.2.6)...
Setting up ettercap-graphical (1:0.8.4.1-2)...
Setting up kali-desktop-core (2026.2.11)...
Setting up kali-desktop-xfce (2026.2.11)...
Processing triggers for ca-certificates-java (20260311)...
done.
Setting up default-jre-headless (2:1.25-77)...
Setting up openjdk-21-jre:amd64 (21.0.11-10-1)...
Setting up openjdk-25-jre:amd64 (25.0.3+9-2)...
Setting up default-jre (2:1.25-77)...
Setting up patator (1.1.0-1)...
Setting up kali-linux-headless (2026.2.11)...
Setting up burpsuite (2026.3.2-0kali1)...
Setting up kali-tools-top10 (2026.2.11)...
Setting up kali-linux-default (2026.2.11)...
Processing triggers for dictionaries-common (1.31.4)...
Processing triggers for ca-certificates (20260223)...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
Processing triggers for initramfs-tools (0.151)...
update-initramfs: Generating /boot/initrd.img-6.19.14-kali-amd64
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.44.6+dfsg-2)...
Processing triggers for php8.4-cgi (8.4.22-1)...
Processing triggers for libapache2-mod-php8.4 (8.4.22-1)...
Processing triggers for libc-bin (2.42-10)...
Processing triggers for ca-certificates-java (20260311)...
done.
kali@kali: ~$
kali@kali: ~$ mkdir -p ~/lab/scans,notes,pcaps,screenshots
kali@kali: ~$
kali@kali: ~$ ls -l ~/lab
notes  pcaps  scans  screenshots
kali@kali: ~$
```

Step 4: Docker Installation

Docker was installed inside Kali Linux to deploy vulnerable applications securely in containers. Docker provides an isolated and lightweight environment for running applications.

Commands Used

```
Shell
sudo apt install -y docker.io
```

To enable Docker services automatically:

```
Shell
sudo systemctl enable --now docker
```

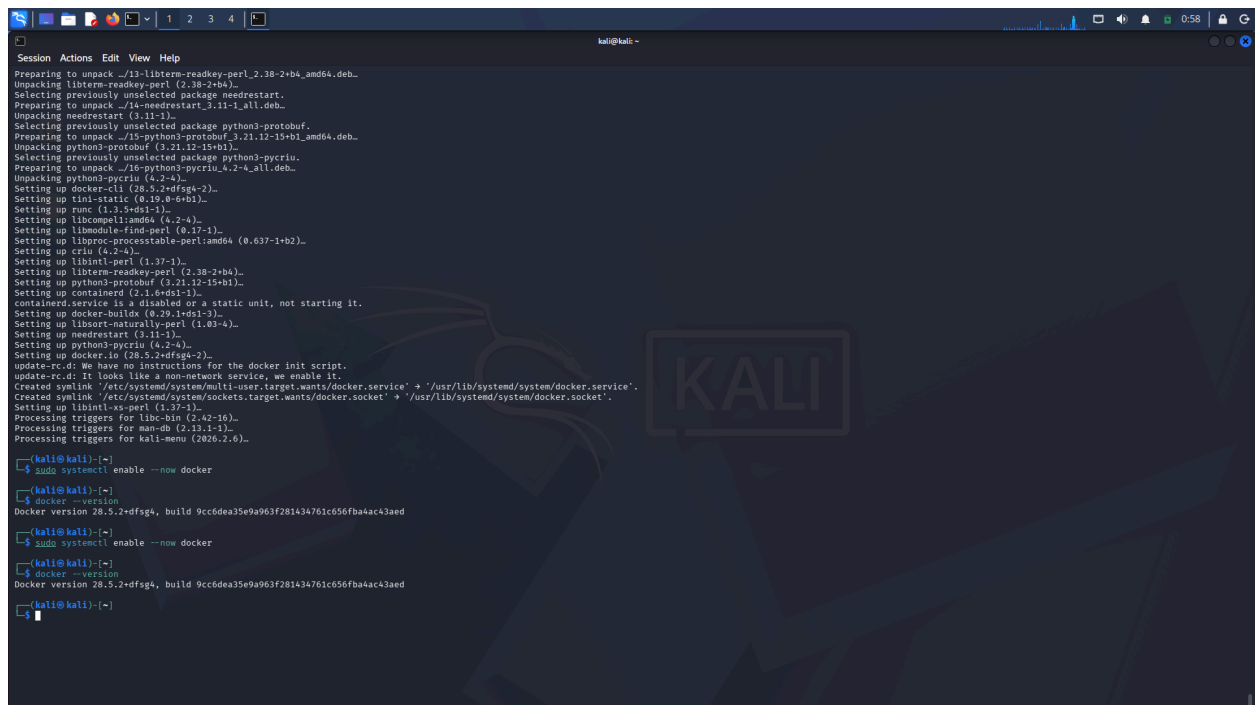
To verify successful Docker installation:

```
Shell
docker --version
```

The successful output confirmed Docker was properly installed and functional.

Screenshot Evidence

Docker Installation Verification



```
kali@kali: ~
Session Actions Edit View Help
Preparing to unpack ./libterm-readkey-perl_2.38-2+b4_amd64.deb...
Unpacking libterm-readkey-perl (2.38-2+b4)...
Selecting previously unselected package needrestart.
Preparing to unpack ./lib-needrestart_3.11-1_all.deb...
Unpacking needrestart (3.11-1)...
Selecting previously unselected package python3-protobuf.
Preparing to unpack ./lib-python3-protobuf_3.21.12-1+b1_amd64.deb...
Unpacking python3-protobuf (3.21.12-1+b1)...
Selecting previously unselected package python3-pycurl.
Preparing to unpack ./lib-python3-pycurl_4.2-4_all.deb...
Unpacking python3-pycurl (4.2-4)...
Setting up docker-cli (20.10.17~ds1-2)...
Setting up tini-static (0.19.0-6+b1)...
Setting up runc (1.1.5-1)...
Setting up libcomptel-amd64 (4.2-4)...
Setting up libmodule-find-perl (0.17-1)...
Setting up libproc-processtable-perl:amd64 (0.637-1+b2)...
Setting up criu (4.2-4)...
Setting up libintl-perl (1.37-1)...
Setting up libterm-readkey-perl (2.38-2+b4)...
Setting up python3-protobuf (3.21.12-1+b1)...
Setting up containerd (2.1.6-1)...
containerd.service is a disabled or a static unit, not starting it.
Setting up docker-buildx (0.20.1-1+b1)...
Setting up libsort-naturally-perl (1.03-4)...
Setting up needrestart (3.11-1)...
Setting up python3-pycurl (4.2-4)...
Setting up docker.io (20.10.17~ds1-2)...
update-rc.d: We have no instructions for the docker init script.
update-rc.d: It looks like a non-network service, we enable it.
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service'.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket'.
Setting up libintl-xs-perl (1.37-1)...
Processing triggers for libc-bin (2.42-10)...
Processing triggers for man-db (2.12.1-1)...
Processing triggers for kali-menu (2020.2.6)...

(kali@kali)~$ sudo systemctl enable --now docker
(kali@kali)~$ docker --version
Docker version 20.10.17, build 10dc92c

(kali@kali)~$ sudo systemctl enable --now docker
(kali@kali)~$ docker --version
Docker version 20.10.17, build 10dc92c

(kali@kali)~$
```

Step 5: Deployment of OWASP Juice Shop

OWASP Juice Shop was deployed as the intentionally vulnerable web application target. It is one of the most widely used applications for practicing web application penetration testing.

Why OWASP Juice Shop?

OWASP Juice Shop provides:

- Vulnerable login forms
- Misconfigured security controls
- Broken authentication testing
- Injection vulnerabilities
- API security testing

It serves as an ideal practice platform for ethical hacking and cybersecurity learning.

Command Used to Pull Juice Shop Image

Shell

```
sudo docker pull bkimminich/juice-shop
```

Command Used to Run Juice Shop

Shell

```
sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
```

Container Verification

The following command was used to verify the running container:

Shell

```
sudo docker ps
```

Successful deployment was confirmed when Juice Shop became accessible at:

<http://localhost:3000>

The vulnerable web application loaded successfully in the browser.

Screenshot Evidence

OWASP Juice Shop Container Running in Docker

```
kali@kali:~$ docker --version
Docker version 20.10.17, build 100c80bf

kali@kali:~$ sudo systemctl enable --now docker

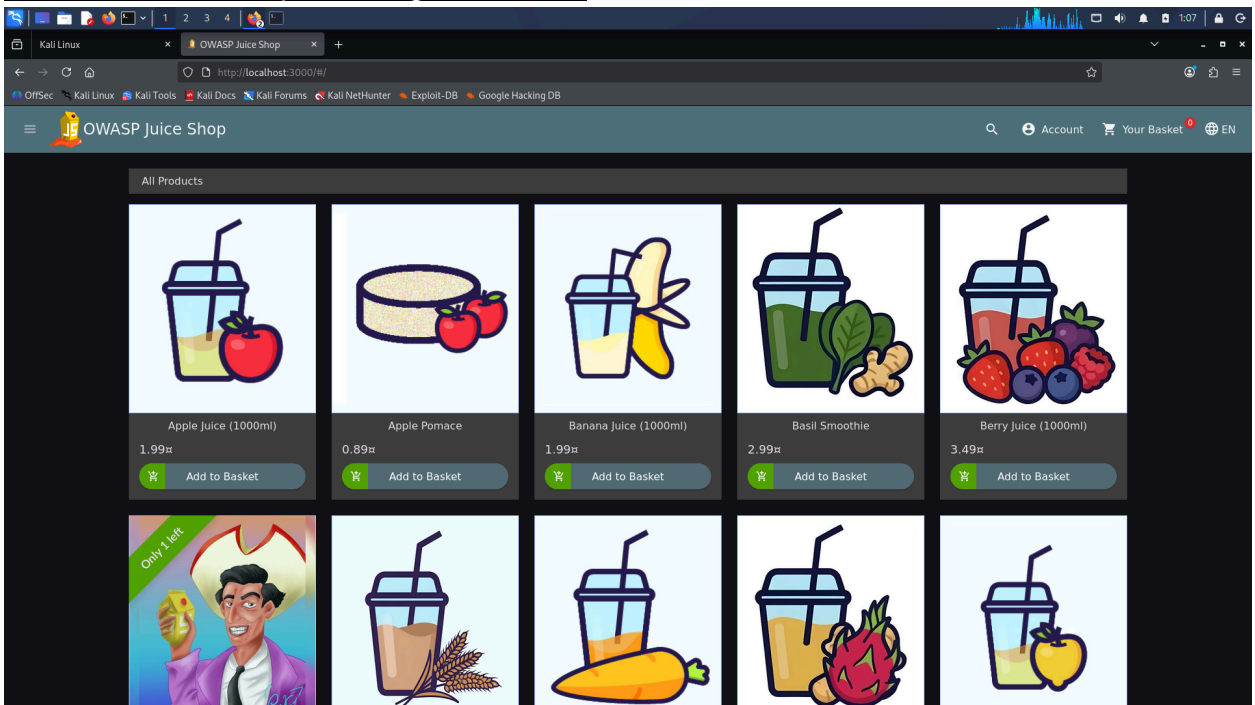
kali@kali:~$ docker --version
Docker version 20.10.17, build 100c80bf

kali@kali:~$ sudo docker pull bkimminich/juice-shop
Using default tag: latest
latest: Pulling from bkimminich/juice-shop
670e5d0b0812: Pull complete
c172f21814df: Pull complete
99515e7b4d35: Pull complete
990a982a9142: Pull complete
d6b1b89eacac: Pull complete
2780928e5dbf: Pull complete
7c128958777b: Pull complete
321ac345c0: Pull complete
52630fc75a18: Pull complete
d6a0b72ad177: Pull complete
b039d9ae01f6: Pull complete
e8ddc55facdc: Pull complete
bdfc7f7a3b6f: Pull complete
2d4d7ad6272: Pull complete
40808157d8d2: Pull complete
b48992a292d1: Pull complete
cac2ae0193cb: Pull complete
d08090f3d3be: Pull complete
7c7b0f68bdf: Pull complete
8224d91da70b: Pull complete
1a4be552d892: Pull complete
39dfe0b31bc: Pull complete
5e22571f0ab1: Pull complete
d08080b4b288: Pull complete
Digest: sha256:5d572829161b32e065edefc9b09ca88567296ff7134ebe3181ab7ac9e9ed4056
Status: Downloaded newer image for bkimminich/juice-shop:latest
docker.io/bkimminich/juice-shop:latest

kali@kali:~$ sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
b70eb441a73a0bf8ce0fee3772a9ac6cabd682e1ac96be957c6b1bcb424b31a

kali@kali:~$ sudo docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS                               NAMES
b70eb441a73a   bkimminich/juice-shop  "/nodejs/bin/node /j..."  12 seconds ago Up 11 seconds  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   juice
```

OWASP Juice Shop Running in Browser



7. NETWORK CONFIGURATION

The cybersecurity lab environment used a dual-network configuration to ensure secure isolation and internet access.

Network Design

Network Type	Purpose
NAT Adapter	Internet Access
Host-Only Adapter	Internal Lab Communication

IP Configuration

The IP address configuration was verified using:

```
Shell
ip a
```

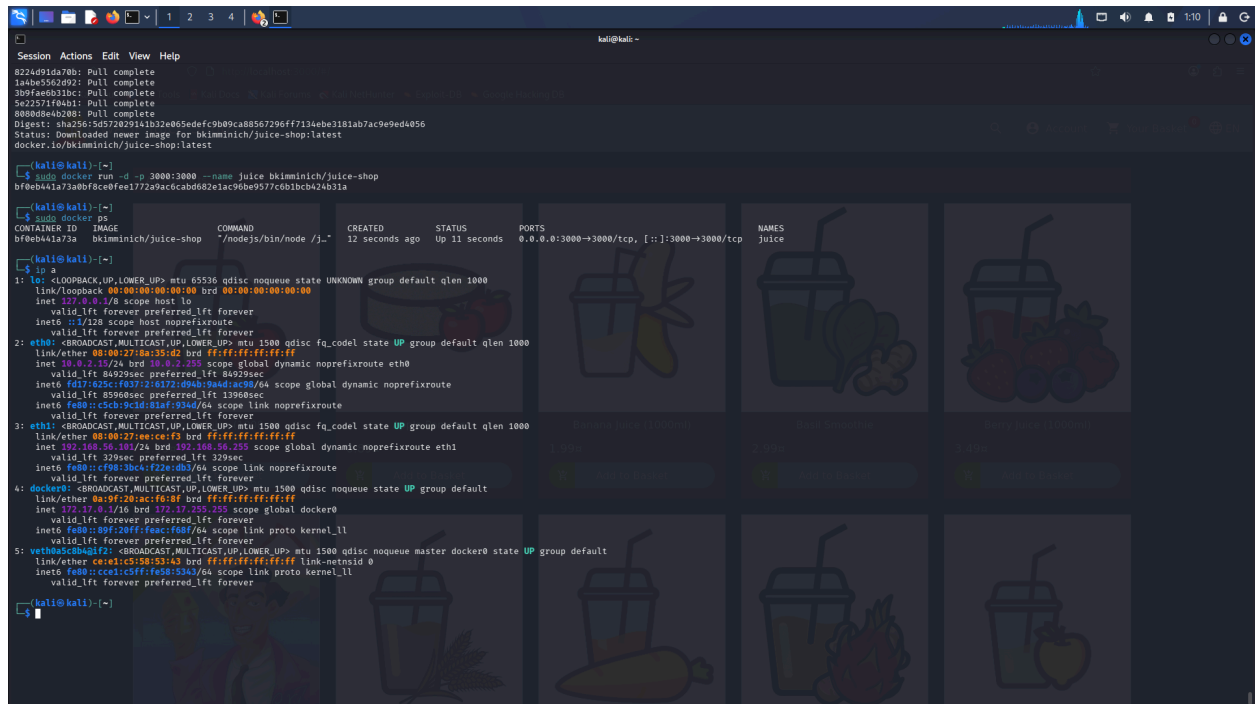
Network Interfaces Observed

Interface	Purpose	IP Address
eth0	NAT Adapter	10.0.2.15
eth1	Host-Only Adapter	192.168.56.101

The Host-Only adapter IP (192.168.56.101) was used for lab testing and validation.

Screenshot Evidence

Network Interface Configuration (ip a)



The screenshot shows a Kali Linux terminal window with the following content:

```
kali@kali:~$ docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
b70eb44a73a8b8f8ce0fee172a9ac6cab0682e1ac90be9577c0b10cb424031a

kali@kali:~$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED    STATUS    PORTS    NAMES
b70eb44a73a8   bkimminich/juice-shop /nodejs/bin/node /j-    12 seconds ago    Up 11 seconds    0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp    juice

kali@kali:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:0a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 86929sec preferred_lft 86929sec
    inet6 fd17:625c:f837:2:6372:d9ab:9a4d:a906/64 scope global dynamic noprefixroute
        valid_lft 85960sec preferred_lft 13968sec
    inet6 fe80::c5c8:9c1d:81af:93a3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:0e:ce:c3 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
        valid_lft 328sec preferred_lft 328sec
    inet6 fe80::c981:3bc4:722e:db3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: dockero: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 0a:9f:20:ac:f6:8f brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global dockero
        valid_lft forever preferred_lft forever
    inet6 fe80::89f:20ff:feac:f68f/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
5: vethpa08b112: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master dockero state UP group default
    link/ether ce:e1:c5:58:53:43 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::ce:e1:c5:58:53:43/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

8. VALIDATION & TESTING

8.1 Nmap Network Validation

Nmap was used to verify network connectivity and discover active devices within the Host-Only network.

Command Used

Shell

```
nmap -sn 192.168.56.0/24
```

Purpose

- Verify Host-Only network communication
- Identify active devices
- Validate isolated lab setup

The scan successfully identified active hosts in the network, confirming that the Host-Only network configuration was functioning correctly.

Screenshot Evidence

Nmap Host Discovery Scan Output

```
(kali㉿kali)-[~]  
$ nmap -sn 192.168.56.0/24  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-14 01:12 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0014s latency).  
MAC Address: 0A:00:27:00:00:32 (Unknown)  
Nmap scan report for 192.168.56.100  
Host is up (0.00054s latency).  
MAC Address: 08:00:27:8A:9C:CD (Oracle VirtualBox virtual NIC)  
Nmap scan report for 192.168.56.101  
Host is up.  
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.04 seconds
```

8.2 Burp Suite Traffic Interception Validation

Burp Suite Community Edition was used to intercept HTTP requests generated by OWASP Juice Shop.

The browser traffic was routed through Burp Proxy using:

Setting	Value
Proxy Address	127.0.0.1
Port	8080

The interception process successfully captured HTTP requests sent to:
http://localhost:3000

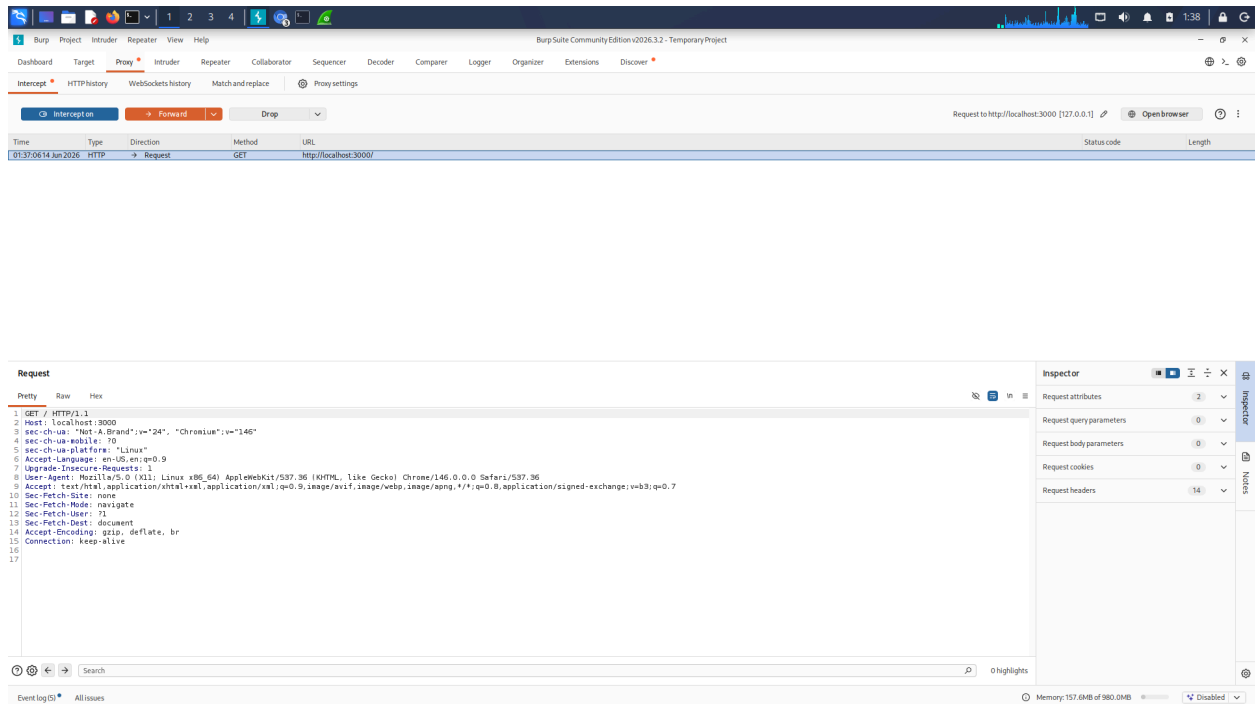
This confirmed successful traffic interception and proxy functionality.

Benefits of Burp Suite

- HTTP request inspection
- Web vulnerability testing
- Traffic modification
- Request replay and analysis

Screenshot Evidence

Burp Suite Intercepting HTTP Request



8.3 Wireshark Packet Capture Validation

Wireshark was used to capture network traffic flowing through the Host-Only adapter.

Interface Used

eth1

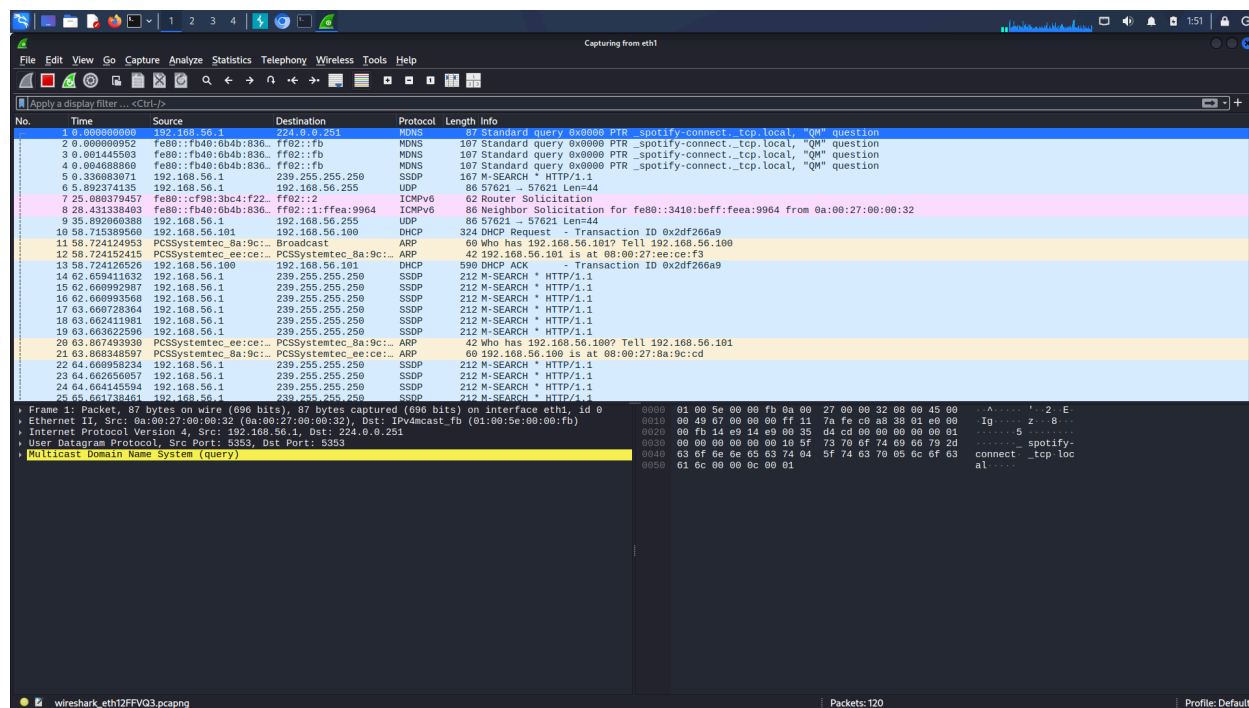
Purpose of Wireshark

- Analyze network packets
- Monitor lab communication
- Observe HTTP traffic
- Validate network communication

Traffic generated from OWASP Juice Shop was captured and analyzed successfully.

Screenshot Evidence

Wireshark Packet Capture on Host-Only Adapter



9. CONCLUSION

The objective of building a personal cybersecurity lab was successfully achieved. A secure and isolated lab environment was created using VirtualBox, Kali Linux, and OWASP Juice Shop.

The implementation included:

- Virtual machine deployment
- Network segmentation using NAT and Host-Only Adapter
- Docker-based vulnerable application deployment
- Traffic interception using Burp Suite
- Packet analysis using Wireshark
- Network validation using Nmap

The cybersecurity lab now provides a safe environment for practicing ethical hacking, vulnerability assessments, penetration testing, and web application security testing without affecting real-world systems.

This lab setup will be beneficial for future cybersecurity learning and practical experimentation.

10. REFERENCES

1. Kali Linux Official Documentation
<https://www.kali.org/docs/>
2. OWASP Juice Shop Documentation
<https://owasp.org/www-project-juice-shop/>
3. VirtualBox Documentation
<https://www.virtualbox.org/manual/>
4. Docker Documentation
<https://docs.docker.com/>
5. Nmap Official Documentation
<https://nmap.org/docs.html>
6. Burp Suite Documentation
<https://portswigger.net/burp/documentation>
7. Wireshark Documentation
<https://www.wireshark.org/docs/>